

Sessional Exam I

Date: 25/02/2026

Time: 2:00- 3:00 PM

Introduction to Data Science and Design Thinking (BCE-O643)

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

- Q1.** Define *Data Science*. Explain the step-by-step approach to learning Data Science. (CO1)
- Q2.** Describe the five key characteristics of data and explain their significance. (CO1)
- Q3.** What are the major stages involved in a Data Science project? Briefly explain each phase. (CO2)
- Q4.** Discuss the major data security issues and challenges encountered in a Data Science project. (CO2)

Section B

- Q1.** Suppose you are the owner of a Vehicle Rental Company aiming to increase revenue in the upcoming semester. Propose a data-driven strategy to achieve this objective. Discuss your approach in terms of different Data Science roles and clearly explain the responsibilities of each role/persona involved. (CO1)
- Q2.** Explain the applications of Data Science across various domains. Describe any five applications in detail. (CO1)

Introduction to Data Science and Design Thinking (BCE-O633)

Sessional Exam II

Date: 22/04/2026

Time: 2:00- 3:00 PM

Note: Section A contains 4 questions of 6 marks each attempt any 2.

Section B contains 2 questions of 8 marks each, attempt any 1.

Section A

S.No	Question	CO	BL
Q1.	What is data Collection? Discuss the different methods for primary Data Collection.	CO3	3
Q2.	What is Data Reduction? Discuss any two Data Reduction Strategies.	CO4	2
Q3.	A dataset has values 4, 8, 12, 16, 20, 24, 28. Construct a box plot by identifying the minimum, Q1, median, Q3, and maximum.	CO3	4
Q4.	What is design thinking? define in your words.	CO4	5

Section B

S.No	Question	CO	BL
Q1.	What are the stages in Design Thinking? elaborate each stage with diagram.	CO3	2
Q2.	A teacher recorded the test scores of 10 students: 55, 60, 65, 70, 70, 75, 80, 85, 90, 95. Calculate the mean, median, mode, and standard deviation.	CO3	3